*Surface Mount*

# Coaxial-Ceramic Resonator Filters and Multiplexers

50Ω DC to 6 GHz

## The Big Deal

- Low insertion loss with excellent power handling
- Passbands up to 6 GHz
- Fractional bandwidth from <1 to 25%
- Low profile designs with min. height of 0.120”
- Excellent temperature stability
- Rugged construction to handle demanding environmental conditions

## Product Overview

Mini-Circuits’ *Coaxial-Ceramic Resonator filters* offer low insertion loss in very small form factors, using ceramic material with high dielectric constant and superior Q factor. Bandpass and bandstop filters,diplexer and multiplexer designs can be constructed using this technology. Low insertion loss combined with excellent power handling makes these filters well suited for transmitter and receiver signal chains. Advanced filter design and construction can achieve stopband width greater than 3x the center frequency as high as 20 GHz.

All our coaxial-ceramic resonator filters are built with rugged construction, qualified to withstand multiple demanding reflow cycles. Custom integrated assembly with LNA in greatly simplifying system integration. They can be realized in small form factors with high-quality, precise machining for applications where size is critical. Excellent repeatability across units is achieved through precise tuning and process control.

## Key Features

| Feature | Advantages |
|---|---|
| Low insertion loss | Low signal loss results in better SNR in signal chain |
| Fast roll-off | Higher selectivity results in better adjacent channel rejection and dynamic range |
| Wide stop band | Wide spur-free stopband results in better receiver sensitivity |
| Excellent power handling | Well suited for transmitter applications |
| Rugged Construction | These filter assemblies have been qualified over a wide range of thermal, mechanical and environmental conditions including withstanding the stress of extensive solder reflow cycles |
| Small Size | Very well suited for high performance applications where size is a constraint. |
| Temperature stability | Very minimal change in electrical performance across temperature makes these filters suitable for a wide range of operating conditions. |

Notes
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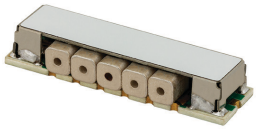
# Bandpass Filter

# CBP-1320Q+

50Ω 1280 to 1360 MHz



*Generic photo used for illustration purposes only*
CASE STYLE: HQ2299

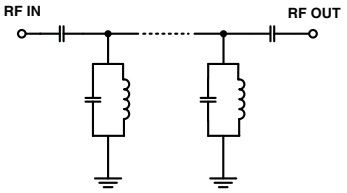
## Features

- Broad stopband performance up to 20 GHz
- High selectivity
- Miniature shielded package

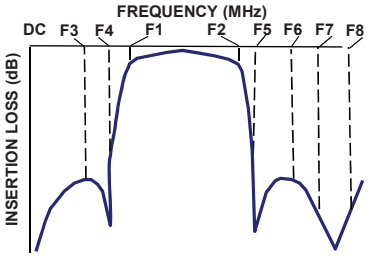
## Applications

- Aviation
- Mobile radio
- Broadband
- Radar and navigation systems

## Functional Schematic




## Typical Frequency Response




***+RoHS Compliant***
*The +Suffix identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications*

## Electrical Specifications at 25°C

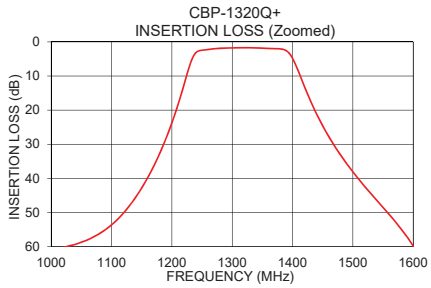
| Parameter | | F# | Frequency (MHz) | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| Pass Band | Center Frequency | - | - | - | 1320 | - | MHz |
| | Insertion Loss | F1-F2 | 1280-1360 | - | 1.9 | 3.0 | dB |
| | VSWR | F1-F2 | 1280-1360 | - | 1.5 | 1.7 | :1 |
| Stop Band, Lower | Insertion Loss | DC-F3 | DC-900 | 50 | 60 | - | dB |
| | | F3-F4 | 900-1170 | 20 | 35 | - | dB |
| | VSWR | DC-F4 | DC-1170 | - | 20 | - | :1 |
| Stop Band, Upper | Insertion Loss | F5-F6 | 1490-1700 | 20 | 30 | - | dB |
| | | F6-F7 | 1700-3000 | 45 | 50 | - | dB |
| | | F7-F8 | 3000-20000 | - | 20 | - | dB |
| | VSWR | F5-F8 | 1490-20000 | - | 8 | - | :1 |

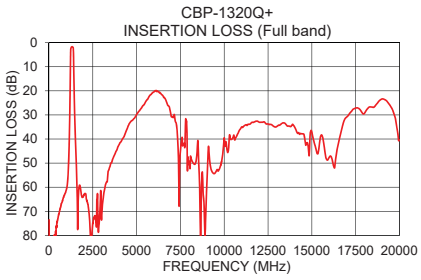
| Maximum Ratings | |
|---|---|
| Operating Temperature | -40°C to 85°C |
| Storage Temperature | -55°C to 100°C |
| RF Power Input | 8 W Max. |

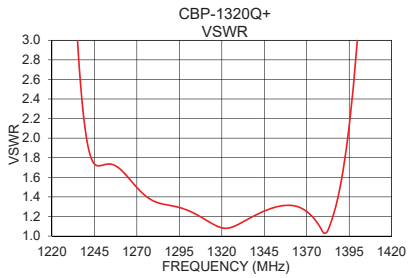
Passband rating , derate linearly to 3.5W at 85°C ambient.
Permanent damage may occur if any of these limits are exceeded.

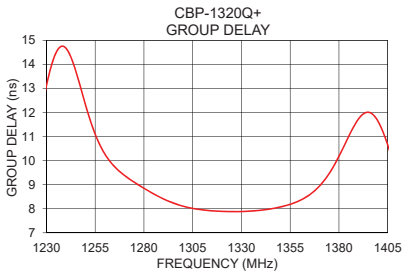
## Typical Performance Data at 25°C

| Frequency (MHz) | Insertion Loss (dB) | VSWR (:1) | Frequency (MHz) | Group Delay (ns) |
|---|---|---|---|---|
| 10 | 73.58 | 442.37 | 1280 | 8.85 |
| 100 | 89.96 | 255.06 | 1284 | 8.66 |
| 900 | 62.98 | 22.46 | 1288 | 8.49 |
| 1170 | 36.39 | 22.08 | 1292 | 8.34 |
| 1180 | 32.63 | 21.79 | 1296 | 8.21 |
| 1210 | 18.51 | 16.41 | 1300 | 8.11 |
| 1240 | 3.13 | 2.04 | 1304 | 8.03 |
| 1280 | 1.87 | 1.36 | 1308 | 7.97 |
| 1320 | 1.70 | 1.08 | 1312 | 7.94 |
| 1360 | 1.92 | 1.31 | 1316 | 7.91 |
| 1400 | 4.80 | 3.04 | 1320 | 7.89 |
| 1440 | 21.17 | 11.05 | 1324 | 7.89 |
| 1470 | 30.62 | 13.15 | 1328 | 7.88 |
| 1490 | 35.75 | 13.91 | 1332 | 7.89 |
| 1700 | 61.12 | 7.55 | 1336 | 7.91 |
| 3000 | 73.63 | 25.27 | 1340 | 7.94 |
| 6075 | 19.94 | 3.80 | 1344 | 7.98 |
| 10000 | 39.52 | 12.64 | 1348 | 8.04 |
| 15000 | 37.22 | 4.96 | 1352 | 8.11 |
| 20000 | 40.47 | 1.27 | 1360 | 8.35 |

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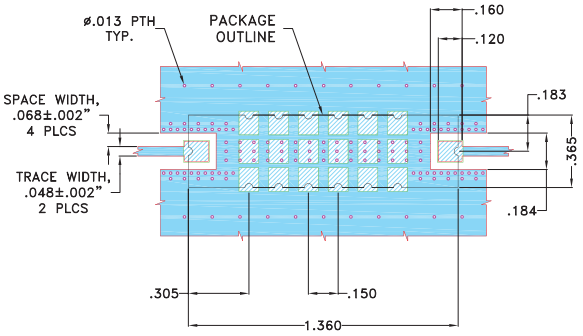
## Pad Connections

| | |
|---|---|
| INPUT | 1 |
| OUTPUT | 8 |
| GROUND | 2,3,4,5,6,7,9,10,11,12,13,14 |

***Demo Board MCL P/N: TB-1006+***
***Suggested PCB Layout (PL-543)***

SUGGESTED MOUNTING CONFIGURATION FOR
HQ2218 & HQ2299 CASE STYLE "14FL01" PIN CODE



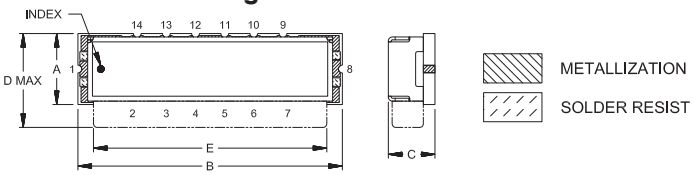

NOTES:

1. TRACE WIDTH IS SHOWN FOR FR4, IT180A WITH DIELECTRIC THICKNESS .025"±.002". COPPER: 1/2 Oz EACH SIDE. FOR OTHER MATERIALS TRACE WIDTH MAY NEED TO BE MODIFIED.
2. BOTTOM SIDE OF THE PCB IS CONTINUOUS GROUND PLANE.

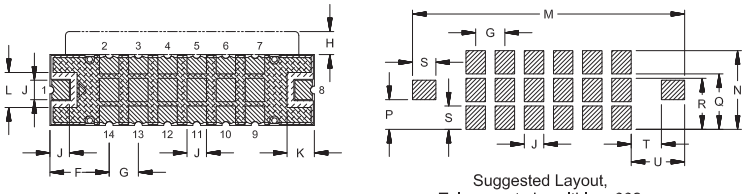
DENOTES PCB COPPER LAYOUT WITH SMOBC (SOLDER MASK OVER BARE COPPER)

DENOTES COPPER LAND PATTERN FREE OF SOLDERMASK

## Outline Drawing




PCB Land Pattern




Suggested Layout,
Tolerance to be within ±.002

## Outline Dimensions ( inch / mm )

| A | B | C Min | C Max | D | E | F | G | H | J | K |
|---|---|---|---|---|---|---|---|---|---|---|
| - | - | | | - | - | - | - | - | - | - |
| **.365** | **1.360** | **.240** | **.270** | **.483** | **1.200** | **.305** | **.150** | **.118** | **.100** | **.140** |
| 9.27 | 34.54 | 6.10 | 6.86 | 12.27 | 30.48 | 7.75 | 3.81 | 3.00 | 2.54 | 3.56 |

| L | M | N | P | Q | R | S | T | U | Wt. |
|---|---|---|---|---|---|---|---|---|---|
| - | - | - | - | - | - | - | - | - | grams |
| **.180** | **1.400** | **.405** | **.153** | **.285** | **.263** | **.120** | **.155** | **.275** | |
| 4.57 | 35.56 | 10.29 | 3.87 | 7.24 | 6.67 | 3.05 | 3.94 | 6.99 | 5.0 |

*Note: Please refer to case style drawing for details*

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